

Course Overview

Stat 151A: Linear Models

- Unit 0
 - Linear algebra
 - * Formal operations: Valid dimensions in matrix multiplication, transpose, trace
 - * Algebra: As a representation of linear systems, invertibility, determinant
 - * Geometry: orthogonal vectors, basis vectors, linear combinations of basis vectors
 - * Eigenvalues and eigenvectors of square symmetric matrices
 - Probability
 - * Means, covariance, variance, correlation
 - * Conditional expectation, conditional variance, independence
 - * Asymptotics: Law of large numbers, central limit theorem, continuous mapping theorem
 - * The standard normal distribution
 - Tasks in data analysis
 - * Descriptive statistics
 - * Prediction
 - * Inference
- Unit 1
 - Simple linear regression
 - * The basic formula
 - * Sample means as a special case
 - * Coefficient as a measure of association
 - * Asymmetry between x and y
 - Matrix form
 - * The regressor matrix and response and residual vectors
 - * The derivation of the formula for vector-valued β
 - * Being able to express familiar simpler regressions in matrix form
 - Including sample mean, simple linear regression

- * ESS, RSS, TSS, and R squared, and the limitations of these measures of fit
- Interpretations
 - * As a descriptive statistic
 - * As a maximum likelihood estimator under normality
 - * As a consistent estimator of a true coefficient
 - * As a risk minimizer for prediction
- Transformations
 - * One-hot encodings and categorical variables
 - * Linear transformations of regressors leave the fit unchanged
 - A one-hot encoding and a constant
 - Rescaling and centering continuous variables
 - * Non-linear transformations of regressors give more expressive regressions
 - Polynomials, indicators, splines
 - * Interactions between regressors and indicator variables
 - Example: different slopes in different categories
 - * Transformations of the response
 - Changing units of the response
 - Choosing the scale that makes the most sense
 - Log transformations of heavy tailed responses
- Unit 2
 -
- Unit 3
 - Expressive basis functions
 - * What are some examples? (Splines, polynomials)
 - * Why do we use them in a prediction context?
 - * What are the tradeoffs determining how many we should use?
 - The bias-variance tradeoff
 - * In the context of linear regression, what does this mean? (Squared error, conditional on the regressors)
 - * Why is it important for prediction problems?
 - * What are some ways to trade off bias for variance in regression problems.
 - Ridge regression
 - * What is it? (Formula, Bayesian interpretation)
 - * How does the ridge parameter affect the bias and variance in the bias-variance tradeoff?
 - * Why do we use ridge regression together with expressive basis functions?

Unit 4 - FWL theorem - Be able to apply it with guidance (as in the homework) - Understand what it means for constants (centering regressors) - Errors in regressors / regression to the mean - Recognizing that inference with OLS assumes that x is measured without noise - If you have noise in your regressors, you underestimate the true effect exactly as in ridge regression - Be able to recognize examples where you're actually trying to measure the effect of something you only observe with noise - Omitted variables - Why do we care for inference? Why don't we care for prediction? - When is omitted variable bias a problem? (When the omitted variable is associated with both the response and variable of interest — apply the FWL theorem) - Understand why you can't get around omitted variables by putting every possible regressor in the regression - Influence and outliers - Whether something is an outlier depends strongly on context - Understand the different kinds of influence of x outliers and y outliers - Know how to diagnose (residuals for y , leverage for x) - Know basic properties of leverage scores (between 0 and 1, sum to P)